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VI-OCR: "Visually Impaired" Optical Character Recognition Pipeline for Text Accessibility Assessment

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ABSTRACT

Low vision adversely impacts daily activities, particularly reading. However, quantifying text accessibility for different levels of low vision is challenging, leading to product designs that often overlook the vision status of low vision readers. In this paper, we bridge the gap between computer vision and low vision fields by introducing a text accessibility assessment pipeline called **VI-OCR** (short for **Visually Impaired Optical Character Recognition**), based on state-of-the-art OCR models. VI-OCR mimics human text recognition ability under specified levels of visual acuity and contrast sensitivity loss, to estimate whether text of a given size would be recognizable for a low vision human reader. We benchmarked specialized OCR models and vision-language models in replicating text recognition performances with visual acuity and contrast sensitivity deficits across three reading tasks: letter acuity using ETDRS charts, word acuity using MNREAD charts, and scene text recognition using complex real-life images. Comparing model performance to that of normal vision participants on degraded texts revealed major issues in some models including limited generalizability across reading tasks, difficulties dealing with severe contrast reduction, and overperforming rather than mimicking human observers. However, robust human-like performance of winning models such as Qwen2.5-VL and GPT supports the feasibility of VI-OCR in assessing text accessibility.

Introduction

In the United States, about 5.7 million people are affected by vision impairment, and this number is expected to double by 2050 due to an aging population¹⁻³. Low vision refers to visual impairment that is not correctable by glasses or contact lenses.

Low-vision individuals rely on their residual sight in their daily lives to read text at different viewing distances, including close-range texts such as medicine labels, books, newspapers, and websites, to middle-range physical scene text such as price tags and room numbers, to distant scene texts such as hospital and street signs. Reading difficulty varies significantly among individuals with different levels of vision impairment, and it is the No.1 reason for an individual with low vision to seek vision rehabilitation care^{4,5}. In this paper, we consider text accessibility in the context of text recognition from digital reading materials to everyday scene texts (i.e., texts with complex colors and fonts in everyday scenes⁶).

In the low vision research field, current efforts to improve text legibility focus on offering easier reading materials, such as increasing font sizes or designing low-vision friendly labels and fonts^{7,8}, developing assistive tools for magnification or enhancing text legibility^{9,10}, and providing alternative non-visual access such as text to speech¹¹. Building managers, signage designers, eye-care clinicians, and family members may wish to evaluate the accessibility of text for a specific level of low vision. However, it is difficult for people with normal vision to subjectively determine what is not legible for a specific level of low vision, and it is often not practical to conduct extensive field tests with individuals with actual low vision. Some researchers have attempted to simulate low vision to assist in the evaluation of text legibility^{12,13}. Among these, digital low-vision filters modeled with contrast sensitivity functions (CSFs)¹⁴ parameterized by clinically available visual acuity (VA) and contrast sensitivity (CS) have achieved good approximations for actual low vision letter and word recognition performance using black on white texts, providing a validated simulation to examine the accessibility of simple texts for various levels of low vision¹⁵.

Simulating user groups has long been a cornerstone of Human-Computer Interaction (HCI), allowing designers to tailor user experiences and conduct usability evaluations at scale. As listed in Table 1, simulation on impaired sensory world using image enhancement or mixed-reality headsets creates a lens through which designers can test inclusive designs and evaluate usability from the target user's perspective¹⁶⁻¹⁸, rehabilitation specialists can experience immersive accessibility

of the assisted population¹⁹, and researchers can collect and observe behavioral patterns through simulated subjects^{20,21}. However, these immersive simulations are primarily for education and empathy building purposes, providing simplified, often single-level simulation of visual acuity or field loss. These simulations are not capable of providing parameterizable options that correspond to specific levels of vision impairment. While most recent work has attempted to incorporate textual prompts for Vision-Language Models (VLMs) to describe characteristics of low vision individuals, these models struggle to imagine perceptual experiences purely from diagnostic or descriptive language^{22,23}. We posit that if low vision characteristics could instead be simulated directly through visual input, rather than through textual description, such visual simulation could narrow the gap between language and perception across a broad range of vision tasks.

In the context of our main focus of text recognition, recent advances in Optical Character Recognition (OCR) have reached human-competitive accuracy in complex texts captured from real world scenes^{24,25}. Given an image, OCR text spotters can simultaneously locate and recognize texts²⁶, enabling many real-world applications such as document digitalization²⁷, intelligent assistants¹¹, and autonomous driving²⁸. Early OCR technology succeeded only under simple conditions, relying on hand-crafted features and rule-based heuristics. However, as large-scale AI-driven applications proliferate, new demands have emerged: OCR engines need to generalize across a broader range of documents, from multilingual content²⁹ to handwritten notes³⁰, and complex layouts including text lines³¹ tables structures³², chart understanding³³, and scene images recognition⁶. Text recognition has also become one of the important evaluation dimensions of the effectiveness of VLM as well³⁴. VLMs are multi-modal models that can learn from images and text input and generate text responses interactively. Large VLMs can generalize well to wide use cases (e.g., chatting, visual question answering, image recognition, etc.) and work with many types of images (documents, scene texts, etc). In this paper, *OCR models* refer to both OCR specialized models and VLMs. To generalize to broader application scenarios from document to scene text, these models are trained with not only horizontally arranged³⁵, machine-printed words³⁶, but also curved³⁷, arbitrarily shaped⁶ and multi-oriented³⁵ texts (e.g., artistic fonts arranged in arcs on billboards). In our case, an additional challenge is aligning reading performance across diverse user groups, which requires OCR engines to be robust not only to document variability but also to differences in visual perception, user interaction, and accessibility needs. While recent OCR development has started to see applications in the population with vision impairment, these efforts primarily focus on text-to-speech applications¹¹. The potential of using OCR for evaluating text visual accessibility has not yet been explored.

These progresses in OCR prompted us to propose further automating the text accessibility evaluation process by combining the low vision simulation method and OCR models to assess text accessibility efficiently, automatically and objectively. Unlike the OCR applications for assisting text recognition, using OCR for text accessibility evaluation in low vision requires the OCR models to mimic, rather than outperform, individuals with low vision. This evaluation task is similar to an image quality assessment task (IQA)³⁸, but traditional IQA focuses solely on quality corruption caused by environmental corruptions such as movement distortion, camera distortion, and compression corruption³⁹ rather than visual impairments.

Figure 1 illustrates the proposed *VI-OCR* pipeline. VI-OCR takes inputs of 1) an image of text with specified angular size and 2) a specific level of low vision quantified by clinically available acuity and contrast sensitivity. The low vision parameters are used to generate corresponding low-vision filters to apply image simulation. The model's performance on recognizing the degraded texts is used to quantify text accessibility for that level of low vision. In other words, the VI-OCR pipeline hypothesizes that **if the OCR model with a specific level of low vision cannot recognize a text with a specified angular size, it is likely that an actual low vision individual with the same acuity and contrast sensitivity would also struggle to recognize the same text when seen under similar geometric conditions** (i.e., with the same angular size).

In this study, we aim to explore the feasibility of text accessibility prediction using VI-OCR pipeline and to evaluate how accurately state-of-the-art OCR models, as the base of VI-OCR, replicate text recognizability of different levels of low vision. In previous works, a critical requirement of valid low-vision simulation is that target features in the original image that are not visible or recognizable with specific levels of low vision are not visible or recognizable to normally sighted subjects viewing the filtered image^{40,41}. These approaches for validating low-vision simulation can be transferred to our current model evaluation. Additional important considerations in our model evaluation include: 1) models should show consistent performance across various reading tasks common in everyday context; 2) models should show robust replications of text recognition across a broad range of low vision; and 3) models need to be feasible for fine-tuning and real-time evaluation.

In Experiment 1, we examined whether VI-OCR could replicate low vision letter and word acuity (the smallest printed letters or words that one can read) when viewing letter acuity and reading acuity charts degraded with low vision filters. These two tasks have been validated by comparing reading performance of participants with simulated low vision to that of real low vision participants¹⁵. In Experiment 2, we conducted exploratory investigation into the feasibility of extending VI-OCR to more complex real-life scene texts, asking whether VI-OCR and normally sighted humans would exhibit comparable performance when recognizing text occurring in complex real-world scenes filtered to represent various levels of low vision.

Experiment 1: VI-OCR performance on replicating low vision letter acuity and word acuity

In this study, we demonstrate the concept of VI-OCR by evaluating whether state-of-the-art OCR models can simulate intended levels of low vision in clinical letter acuity and word acuity tasks. Each CSF-based low-vision filter corresponds to an intended goal level of low vision acuity, named *Expected Acuity*. Prior study¹⁵ have demonstrated that such filters can closely replicate the intended reductions in letter and word acuities in human participants with simulated low vision. Proving that VI-OCR exhibits a comparable sensitivity to reduced acuity and contrast sensitivity as normally-sighted human observers, for both letter acuity and word acuity, is the first step in validating the VI-OCR pipeline.

Methods

Low Vision Conditions

Low-vision filters were generated by shifting a normal-vision CSF template S_{NV} downward and leftward to reduce the peak contrast sensitivity and high spatial frequency cutoff (Figure 2). The normal-vision CSF characterizes a normally sighted individual's acuity limit and contrast sensitivity across a range of spatial frequencies⁴² based on Barten's simplified CSF formula⁴³. The two shifting parameters (a for horizontal shift and b for vertical shift) were chosen based on the goal level of visual acuity and contrast sensitivity that the filter attempts to simulate (Eq 1). More details about the low-vision filters and the derivation of a and b were described in Xiong et al. 2021¹⁵ and Thompson et al. 2017⁴⁰. The CSF filters were then constructed by computing the spatial-frequency-specific attenuation when comparing the low vision CSF and the normal CSF template (Eq 2). To apply a CSF filter to any digital or physical image with known viewing angle, the image first underwent Fourier transformation to obtain its amplitude at each spatial frequency, which was then multiplied by the corresponding value of the CSF filter function to achieve spatial-frequency specific attenuation. Lastly, an inverse Fourier transform was applied to create the filtered image. Figure 1 illustrates the effects of three filters on three example images.

$$S_{LV}(f) = \min\left(b \cdot S_{NV}\left(\frac{f}{a}\right), S_{NV}(f)\right) \quad (1)$$

$$F(f) = \frac{S_{LV}(f)}{S_{NV}(f)} \quad (2)$$

where S_{NV} and S_{LV} are contrast sensitivity of normal vision and simulated low vision at spatial frequency f , F is the CSF filter constructed by computing the attenuation ratio between the normal and low vision CSF at each spatial frequency component.

To examine the generalization of VI-OCR performance in simulating different characteristics of low vision, we chose 15 low vision conditions including 5 conditions with horizontal shifts primarily leading to high spatial frequency reduction, 5 conditions with vertical shifts primarily leading to peak contrast sensitivity reduction, and 5 conditions with combined horizontal and vertical shifts of the normal vision CSF template. Table 2 lists the a and b parameters for the 15 low-vision filters, and the expected visual acuity associated with each filter. Note that a leftward shift of normal vision CSF is only associated with VA loss but not CS loss, while a downward shift is associated with both CS and VA loss. Therefore, we used VA as the primary outcome for model evaluation in this experiment as it is sensitive to any combinations of leftward and downward shift. Using 0 logMAR as an average normal vision baseline, the expected acuity deterioration ranged from 0.6 logMAR to 1.8 logMAR across all 15 conditions.

OCR Model Selection

We selected 22 open-source and closed-source OCR specialized models, VLMs and a widely used mobile app, SeeingAI, in this experiment. Closed-source models were previously reported to outperform open-source models due to proprietary training data and advanced post correction techniques⁴⁴, but their internal architectures are often not accessible which prevents further development or customization of the model. We included these models in our current evaluation to see whether current state-of-the-art closed-source models remain to outperform open-source models, and if so, how much difference exists.

We selected three OCR specialized models, including a closed-source model Google Vision⁴⁵, and two open-source models DBNet++ & MAERec (a detection + recognition model^{25,46}) and SPTSv2 (an end-to-end model⁴⁷). Six state-of-the-art VLM series were selected, including closed-source models: GPT Series (GPT4o, 4o-mini, 5, 5-mini⁴⁸), Gemini Series (Gemini 2.5 Flash, 2.5 Pro and 2 Flash⁴⁹), Claude Series (Claude 3.5 Haiku and 3.7 Sonnet⁵⁰), as well as open-source models: Deepseek-OCR Series (Tiny, Small, Base, Large, Gundam)⁵¹, Qwen2.5-VL Series (Qwen2.5-VL 3B, 7B, 32B⁵²) and CogAgent⁵³. In addition, we included a commercially available software SeeingAI¹¹, an app designed for assisting blind and visually impaired individuals that offers OCR functions through image uploading. We excluded another widely used app, BeMyAI⁵⁴, as it only supports real-time camera capture, which could introduce additional camera distortions (e.g., variations in instantaneous field of view and resolution). More details on the model selection are provided in Appendix 1.

When reporting the results, we provide figures for the most human-like VLMs and all OCR-specialized models. The complete results of all models are provided in the Appendix.

Tasks and Scoring

Sixteen digital versions of ETDRS⁵⁵ and 16 digital versions of MNREAD⁵⁶ charts generated in Xiong et al., 2021 were used in this experiment.

An ETDRS chart had 20 lines, with the x-height of the letters ranging from 248 to 3 pixels in steps of 0.1 log unit. A MNREAD chart consisted of 13 lines, with the x-height ranging from 67 to 4 pixels in steps of 0.1 log unit. Each ETDRS or MNREAD chart corresponded to one of the 16 conditions (baseline without filtering and 15 filter conditions) and was submitted to each of the candidate OCR models for letter or word recognition tasks. Example images are provided in Figure 2.

We followed the standard clinical scoring rules in calculating the letter⁵⁵ and word acuities¹⁵ for OCR models under each low-vision filter, as well as under the baseline condition without the filters. Specifically, we calculated letter or word acuity as the print size of the smallest line on which at least one letter or word can be recognized correctly, and adjusted by letter or word errors. Each letter on the ETDRS chart was worth 0.02 log unit, and each word on the MNREAD chart was worth 0.01 log unit.

For the baseline condition without any filtering, an OCR model's "acuity", i.e. the smallest letter that can be identified correctly, is determined by the resolution limit of each model based on their intrinsic model structures. We computed this resolution limit using the x-height unit^{57,58} widely used in the typography field, i.e., number of pixels constructing the height of a lower case "x". Because our primary interest was in how much the model's acuity would change across the filtering conditions and how these changes would compare with corresponding changes in human performance, model acuity under the no-filter condition was assigned a nominal acuity of 0 log unit (in analogy to the 0 logMAR for human observers) to account for the model difference in their resolution limits.

Data Analysis

Reading performance from sixteen normally sighted human observers under the same 16 filter conditions were obtained from Xiong et al., 2021 for comparison.

To reflect sensitivity to the reduction in VA and CS across models and human, we performed the statistical analyses on **Acuity Change** in log units, calculated as the difference between the acuity under each filter condition in relation to the no-filter condition. We further fitted linear regression models to examine whether the OCR models demonstrated the expected acuity changes across all filter conditions. Expected acuity change refers to the difference of acuity between original and low-vision conditions using expected acuity. We considered ± 0.2 log unit as clinically significant difference in acuity change⁵⁹. To assess agreement between model- and human-measured reading performance as vision worsened, we performed Bland–Altman (BA) analysis⁶⁰. In a BA plot, the x-axis represents the mean visual acuity values between the model and human performances, while the y-axis (bias, in logMAR) quantifies the deviation between model and human performances.

To validate the simulation effect, we used two sets of Root Mean Square Error (RMSE) to quantify the overall absolute difference 1) $RMSE_{exp}$ between the model performances and expected acuity reduction, i.e., how well a model or human reaches expected acuity reduction; and 2) $RMSE_{human}$ between model acuity reduction and human acuity reduction, i.e, how closely a model resembles actual human performance.

$$RMSE_{exp} = \sqrt{\frac{\sum_{i=1}^N (Acuity_i - \hat{Acuity}_i)^2}{N}} \quad (3)$$

$$RMSE_{human} = \sqrt{\frac{\sum_{j=1}^N (Acuity_j - \tilde{Acuity}_j)^2}{N}} \quad (4)$$

where N is number of filters. In $RMSE_{exp}$, $\hat{Acuity}_i$ is the expected acuity change under filter i , and $Acuity_i$ represents the model or human measured acuity change under the same filter i , respectively. In $RMSE_{human}$, $\tilde{Acuity}_j$ is the human-measured acuity change under filter j , and $Acuity_j$ is the model acuity change under the corresponding filter.

Results

Letter Acuity

Figure 3 (A)-Human replots the comparison reported in Xiong et al., 2021 between the actual acuity change of human participants with normal vision observing the filtered charts and the expected acuity change associated with the low-vision filters. Human observers with normal vision showed acuity reductions that closely approximated the expected effect ($R^2 = 0.92, p < 0.001$).

Under the no-filter condition, the models displayed large differences in their resolution limits for letter recognition, ranging from x-heights of 4 pixels for DBNet++ & MAERec, SeeingAI and GPT Series to 82 pixels for SPTSv2. (Appendix 2).

Figure 3 (A) illustrates the changes in model acuity (in log units) as a function of the expected changes (in logMAR) under the filtered conditions. We report results from 3 OCR-specialized models, SeeingAI, and the top-7 VLMs selected based on their alignment with the expected acuity change. Results for all evaluated models are provided in Appendix 6. Most models closely replicated the expected low vision acuity with majority of the points landing on or close to the equality line except two OCR-specialized models, SPTS v2 and Google vision. Using 0.2 log unit as a clinically significant difference (shaded gray region in Figure 3 (A)), SeeingAI, GPT4o Mini, GPT 5, GPT 5 Mini and Qwen2.5-VL 32B had almost all points

within this range. The BA plot (Figure 3 (B)) further revealed that most disagreements arose from overperformance of models particularly under severe spatial frequency reduction (high logMAR), where model-measured logMAR values were lower than human-measured reading performance. In Figure 3 (C), the left panel summaries $RMSE_{exp}$ that quantifies the differences between expected acuity changes and measured acuity changes, and the right panel quantifies the differences between the models changes and human acuity changes. The best five models, SeeingAI, GPT5, 5 Mini, 4o Mini and Qwen2.5-VL 32B, closely approximated expected acuity changes under the simulated low vision conditions (Figure 3B, left panel). Meanwhile, Qwen2.5-VL 32B most closely replicated human acuity change (Figure 3 (C), right panel).

Word Acuity

For MNREAD (Figure 4), again human observers with normal vision showed word acuity deterioration that closely resembled the expected low vision word acuity (Figure 4 (A)-Human, replotted from Xiong et al., 2021, $R^2 = 0.92, p < 0.001$).

Under the baseline no-filter condition, the models showed similar resolution limits for word recognition, ranging from x-heights of 3 pixels for GPT4o and Gemini-2.5 Pro to 9 pixels for DBNet++ & MAERec (Appendix 2). The better word acuity than letter acuity for most models aligns with typical observations in human participants.

Under the filtered conditions, all models reached their plateau when viewing charts that simulate severe acuity deterioration beyond 1.2 logMAR. This plateau resulted from the limited range of print size on the MNREAD charts (-0.3 log unit to 0.9 log unit), which constrained text accessibility testing for acuity beyond this range. For acuity deterioration up to 1.2 logMAR, the models consistently replicated the expected pattern of low-vision reading acuity. Using 0.2 log unit as clinically significant difference (shaded gray area in Figure 4 (A)), all selected models except Google Vision and DBNet++ & MAERec maintained small deviations from expected acuity change near or within this criterion. Unlike letter acuity, Figure 4 (B) reveals that overperformance of model increased as visual degradation became more severe across all three types of visual reduction, especially for contrast sensitivity reduction. Figure 4 (C) shows that Qwen2.5-VL, GPT, and the Gemini 2.5 series were the best models for replicating expected low-vision reading acuity, while SeeingAI and Qwen2.5-VL series most closely matched human reading performance. These winning models were largely consistent between the letter recognition and word recognition tasks.

Experiment 2: VI-OCR performance on complex scene texts

In Experiment 2, we extended the evaluation of VI-OCR models to complex scene texts that are often irregular in font types, shapes, contrasts, etc, but most often seen in our everyday life. Using the same low vision simulation method, we examined the changes in scene text recognition models in the VI-OCR framework. Because we do not yet have actual performance from low vision individuals, we focused on comparing the performance of VI-OCR models and normally sighted human observers in scene texts degraded to low vision levels to establish the premise of the VI-OCR framework.

Method

Human Participants

Twenty-three participants with normal vision participated in this study. This study was approved by the Johns Hopkins University Institutional Review Board (IRB00446365) and followed the Declaration of Helsinki. Informed Consent was obtained from each participant prior to their participation.

Model Selection & Low Vision Conditions

Models and low vision conditions were identical to those in Experiment 1.

Task

OCR models and human participants were prompted to recognize texts in images selected from a publicly available scene text dataset TotalText⁶, which primarily includes curved English texts and digits (e.g. "24 hours") annotated with polygonal bounding boxes at the word level.

Twenty scene text images were selected from the test set of TotalText that showed the best baseline performance (F1 score > 0.75 , definition of F1 see below) across the twenty-two OCR models in a preliminary evaluation. This selection criterion was to avoid a floor effect of OCR performance under low vision conditions. Model evaluation result using broader set of images is provided in Appendix 5. Figure 5 shows the 20 images used in this study. The images were presented on a 13.3-inch laptop with a resolution of 2560 by 1600 pixels. The height of the letters ranged from 4 to 195 pixels, corresponding to angular sizes of 0.06 deg (-0.09 logMAR) to 3.13 deg (1.59 logMAR) at a viewing distance of 40 cm. All 20 images were filtered to each of the 15 low vision conditions, totaling 320 text images, including the non-filtered images. Most scene text images have multiple texts, totaling 110 text items for each filter, and 1760 text items across all filters.

The human data collection was performed using a customized image annotation protocol powered by LabelMe⁶¹. The human participants were asked to perform text recognition across the 20 images, with the low vision condition for each image

randomized across participants. Participants were asked to both detect the texts by drawing polygons around the texts and recognize the texts by typing in a textbox. This current report focuses on the recognition results. For OCR evaluation, each of the text images was submitted to each OCR model once with all 16 conditions (no filter and 15 filters). We fine-tuned End-to-end model (SPTS v2) and the detector of two-stage model (DBNet++) on training set of TotalText, while VLMs and closed-source models were evaluated under zero-shot.

Data Analysis

We quantified both human and model scene text recognition performance by a word-level F1 score for each filter condition and the no-filter condition. The F1 score (Eq. 5) was computed as the harmonic mean between the recognition precision (Eq. 6) and recall rate (Eq. 7) under each filter condition (all case insensitive). Irrelevant characters (e.g. '-' in 24-hours) in results were rinsed out before evaluation using full lexicon containing all words in The test set⁶².

$$F1_i = 2 \times \frac{Precision_i \times Recall_i}{Precision_i + Recall_i} \quad (5)$$

$$Precision_i = \frac{\text{Correctly Recognized Words}_i}{\text{Total Recognized Words}_i} \quad (6)$$

$$Recall_i = \frac{\text{Correctly Recognized Words}_i}{\text{Total GroundTruth Words}_i} \quad (7)$$

$$RMSE_j = \sqrt{\frac{\sum_{i=1}^N (F1_i^{model} - F1_i^{human})^2}{N}} \quad (8)$$

We calculated Root Mean Square Error ($RMSE_j$) between model j and normally sighted human F1 score as an indication of how "human-like" a model was in reacting to the low vision simulation, where N denotes the number of filter conditions (Eq. 8). One RMSE was obtained for each model across all filter conditions.

Results

Figure 6 (A) shows the average F1 score of normal-vision human observers and models under each of the low vision conditions. Table 3 summarizes the F1 scores and independent t-test across the high spatial frequency reduction, peak contrast sensitivity reduction, and combined reduction conditions. P-values were adjusted for multiple comparisons using the Holm-Bonferroni correction to control the family-wise error rate.

Figure 6 (A) shows that the impacts of the filters were prominent on both human and model F1 scores, which decreased smoothly with more severe low vision simulation. Under the baseline no-filter condition, Qwen series, GPT series, and Claude3.7 Sonnet outperformed the average human F1 score of 0.934. Notably, GPT5 and Qwen2.5-VL 32B achieved the highest average F1 score of 0.953. VLMs outperformed OCR-specialized models not only under simulated low-vision conditions but also in the original unfiltered condition. Although closed-source models generalized better under simulated low-vision conditions, open-source models demonstrated competitive F1 scores under the no-filter condition. In addition, both Qwen2.5-VL and DeepSeek-OCR showed improved accuracy and generalizability to low-vision conditions as model size scaled up. The most significant differences between model and human performance occurred under peak contrast-sensitivity reduction and combined reduction conditions.

We asked how "human-like" the models were by comparing the model F1 scores with the human F1 scores. As shown in the scatter plots in Figure 6 (B), overall, most VLMs and OCR-specialized models closely approximated human participants ($R^2 > 0.731, p \leq 0.001$), with data points falling on or close to the equity line. The BA plots indicated good agreement between the model and human performance especially for VLMs. The selected model with the lowest bias, Qwen2.5-VL 3B, achieved a mean bias of 0.02 (1.96 SD = 0.11). Most samples (95%) fell within these limits, suggesting high consistency and no systematic bias. When the models and human differed, the directions of the deviation varied across models. Specially, under simulations of severe low vision, DBNet++ & MAERec, SPTSv2, and CogAgent had lower F1 scores than humans, indicating poorer performance, while Google Vision, SeeingAI, Gemini and GPT series had higher F1 scores, overperforming human observers. (Figure 6 (C)). Overall, Qwen, Gemini, and GPT series were the most human-like models in this task with the lowest RMSE between the model and human F1 scores (Figure 6 (D)).

Discussion

To quantify text accessibility objectively and efficiently, we proposed a novel VI-OCR pipeline that combines human CSF-based data augmentation with OCR models to simulate low-vision text recognition performance. We empirically benchmarked a wide range of OCR models. Our findings suggest that changes in VI-OCR performance can serve as a proxy for assessing reading accessibility in standardized text, as demonstrated in Experiment 1. While the comparable sensitivity to acuity reduction observed in real-world text in Experiment 2 indicates the potential of VI-OCR as a tool for evaluating text accessibility beyond controlled conditions, further validation with actual low-vision participants will be essential to establish its practical applicability.

Traditional image augmentation techniques vs. vision-loss inspired degradation Two traditional computer vision tasks, image quality assessment and text spotting, have relied on data augmentation techniques to mitigate distribution shifts across natural image variations. Common corruption simulation approaches⁶³ like JPEG compression, spatial transformations, Gaussian noise, and motion blur are widely used to simulate warped, under-perspective, blurry texts. Contrast and brightness are also applied to simulate camera distortions^{64–66}. However, these models all target normal-vision users through considering quality damage posed by natural scenes, motions, and cameras (environmental corruption). None of these augmentation methods accounts for image distortions arising from impaired vision.

While all of the OCR models evaluated here have been trained with datasets with external distortions using normally sighted human perception as ground truth, their effectiveness in handling image degradation due to vision impairments was unclear prior to our study. Here we found that these models showed surprisingly similar trends of performance deterioration under the low vision simulation, when compared to human observers with normal vision viewing the same filtered images. This convergence suggests that OCR models may have implicitly adopted features of human contrast sensitivity functions, as their training data are largely labeled by humans with normal vision under different levels and types of real-world corruption. Some environmental corruptions can produce similar effects to visual reduction, such as overexposure in cameras and contrast sensitivity loss. The overall consistency in model and human performance when dealing with image degradations simulating low vision proves the premise of the VI-OCR text accessibility evaluation framework that models with degraded input images has potential to serve as an approximation of actual people with low vision.

VLMs cannot imagine what they see through textual descriptions of low vision Recent advances in Large Language Models (LLMs) extend perception simulation to role-playing, where a model adopts predefined description of personas as a textual prompt such as an "AI Patient"⁶⁷, "elderly learner", and in our case, "visually impaired reader"^{22,23}. LLMs offer an opportunity to interactively scale accessibility assessment²³, generate synthetic data that reflect underrepresented user experiences, and provide continuous, adaptive feedback⁶⁸. However, current foundation models still struggle to authentically simulate users with sensory disabilities. When representing visual impairments, textual descriptions of visual reduction result in a noticeable misalignment between what the model "imagines" and what a low vision individual actually perceives^{22,23}. We argue that this gap arises from two intertwined limitations: the scarcity of representative training data describing sensory experiences⁶⁹, and the intrinsic difficulty for LLMs to render textual narratives into visual perception. This naturally leads to our research question in this paper: if existing vision models can interpret visual input comparably to normally sighted individuals, can they generalize that ability to reading materials under simulated low-vision conditions—mirroring how designers experience visual degradation through perceptual filters? We conducted an additional experiment comparing the performance of VLMs across two conditions: (1) a visual persona condition, where models received degraded visual input following the VI-OCR framework, and (2) a textual persona condition, where models received normal visual input paired with a textual prompt such as "You are a low-vision reader with a visual acuity of 0.36 logMAR and a contrast sensitivity of 1.59 logCS." As shown in Figure 7, the textual persona failed to reduce the VLM's performance to approximate what would be expected from low vision individuals. Therefore, creating personas of low-vision individuals directly through visual input could extend this approach beyond to purely vision models more reliably, not only the ones used in OCR, but also has potential in semantic segmentation, object detection, and more vision tasks.

Selecting OCR models for VI-OCR framework In real-life VI-OCR application, the texts being evaluated can range from simple letter, printed words, to complex scene texts. The poor performance ($RMSE > 0.5$) of Deepseek-OCR, CogAgent, Google Vision and SPTSv2 with single letters, as well as the large deviations in their acuity changes under simulated low vision exclude them from being considered as candidate OCR models. This poor performance with single letters may be attributed to the models' stronger response to semantic context, which is termed as contextless challenge in OCR modeling²⁵. Contextless text refers to non-semantic text, such as random combinations of letters, digits, or symbols that do not appear in dictionaries. As shown in Figure 8, models may fail to recognize contextless text even when presented with a clear background and minimal distortion. This issue can arise from the over-introduction of semantic information in both the model design and training set corpus, which is also known as vocabulary reliance^{70,71}. Models tend to predict text that aligns with familiar syntax rules from their training data (e.g., mistaking a random ordered letters 'KCDZV' for a semantically meaningful word 'GOODY' in ETDRS) or texts that occur more often in training set (e.g., mistaking "Z" for "N"). This context-less challenge for OCR models shares similarity with human vision as both normally-sighted and low-vision participants exhibit consistent better reading acuity than letter acuity^{8,72–74}. On the other hand, recent advances in robust Visual Question Answering (VQA) modeling have demonstrated the importance of semantic alignment and feature robustness to generate correct responses based on multi-modal inputs^{75–79}. In our case, visual degradations might exacerbate semantic misalignment because corrupted inputs can impair visual feature extraction and weaken vision–language–semantic alignment, thereby increasing reliance on textual cues or low-salience visual signals. Such overreliance, in turn, can elevate the likelihood of guessing within high-frequency, semantically generic words, therefore yielding a skewed answer distribution.

The remaining OCR specialized models, VLMs except Claude series and SeeingAI, all showed good performance

approaching expected acuity changes for letter and word tasks ($RMSE < 0.5$), and were acceptably "human-like" in the scene text task ($R^2 > 0.8, p < 0.001$). However, these models still show greater departures from human performance under more severe low vision conditions, likely due to the lack of low contrast images in these model's training set, or outperformed human acuity (e.g. GPT series), likely due to the lack of data annotated by low vision individuals. To train models that align with actual low-vision performance, these models need to learn to collect visual information from degraded vision inputs in ways similar to real low vision individuals. Therefore, such limitations can be addressed by finetuning these models using unfiltered text images annotated by actual low vision individuals. The need for finetuning makes the close-source models a less ideal candidate.

Comparing all model evaluated, we consider Qwen2.5-VL series and DBNet++ & MAERec to be viable options for VI-OCR, due to their reliable performance across three tasks and being open-source. Further consideration in model selection comes from the uniqueness of low vision dataset. We anticipate that collecting large amounts of low-vision data would be more difficult than collecting data from normal vision, which would limit the scale of candidate models that we could consider as the base of VI-OCR. For instance, a VLM with 2 billion parameters needs at least 600 images to finetune for each medical image modality diagnosis^{80,81}. In our case, we estimate that 1000 well annotated images across different visual reductions are needed to fine tune a three billion VLM for reduced acuity and contrast sensitivity. In contrast, open-source OCR specialized models such as the DBNet++ & MAERec model that we evaluated, though less generalizable in scene text spotting than larger models, are lightweight and require far less training data. In Appendix 3, we demonstrate our small-scale fine-tuning experiments for DBNet++ & MAERec. To consistently and robustly generalize models for low vision conditions, high-quality perceptual annotation and dataset refinement are essential, at least large enough for such lightweight models.

Limitations and future work The low vision CSF used in the VI-OCR framework has been validated by comparing the performance of participants with normal vision reading filtered clinical charts with that of people with low vision at the corresponding vision level. However, this validation has not been fully extended to scene-text scenarios, which involve multiple colors and complex environmental distortions. Unlike the black-and-white, horizontally arranged visual acuity charts, the accessibility of actual scene text is influenced by additional factors such as color, contrast, and orientation, which can vary significantly even under similar low vision conditions. Actual VI-OCR application in scene text introduces further complexity because its baseline performance depends jointly on image resolution and the base OCR model's "acuity". In this study we took existing scene text dataset without resolution considerations from a specific camera. Further validation of VI-OCR should consider both OCR and camera properties when evaluating actual texts in the everyday environment.

The state-of-the-art models and assistive software evaluated in our study already showed promising performance as the base of our VI-OCR pipeline, but they still lack generalizability across OCR tasks and perform poorly with severe low vision simulations. Addressing these challenges requires fine-tuning models on relevant datasets with ground truth performance data provided by actual low vision observers. Beyond standard document and scene text recognition, receipt and check reading are other two important activities for low vision individuals that warrant further evaluation. These tasks introduce challenges involving varied formats, noisy backgrounds, handwritten elements, and the need for fine-grained structural understanding. Evaluating VI-OCR performance in these domains is worthwhile for advancing usability in real-world assistive technologies in the future. We are initiating a low vision accessibility initiative to collect and explore low vision data across a wide range of contexts under different viewing distances and lighting conditions, such as recipes, labels and digital texts often read at a near viewing distance; indoor navigation signages viewed at a median viewing distance; and road maps or signs of restaurants seen from a far distance. Such a low vision specific dataset would facilitate more robust model training and evaluation, ultimately improving the models' reliability and accuracy in real-world applications.

Conclusion

We introduced VI-OCR, a text accessibility assessment framework that can be customized to accommodate the reading capabilities of people with varying degrees of vision loss by adding an image degradation module prior to text spotting. We extended vision tests from traditional black-and-white digital charts to colorful, arbitrarily shaped scene text. State-of-the-art OCR models showed promising performance in replicating low vision letter and word acuities, as well as in approximating human performance in scene text recognition of low-vision filtered real-life scene images through such visual personas, demonstrating VI-OCR as a promising concept for evaluating text accessibility. To better simulate human sensitivity to visual cues, low vision specific datasets should be generated and used to refine these models that reflect the specific text recognition difficulties faced by this population in their activities of daily living.

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Data Availability: The data sets used and /or analyzed during the current study are available from the corresponding author on reasonable request.

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Table 1. Overview of simulation paradigms for low-vision research, illustrating the shift from perceptual and behavioral modeling to scalable, persona-based simulation enabled by large models.

Paradigm	Mechanism	Application	Input	Output
Physical / Mixed-reality Simulation	Uses image processing, AR/VR, or optical filters to emulate impaired sensory inputs.	Design evaluation ^{16,17,82} Rehabilitation training ^{19,20}	Raw image/ scene + visual filters	Subjective assessment
Behavior Simulation	Models user interactions and performance metrics (e.g., reaction time, accuracy) under synthetic impairments.	Accessibility assessment ^{18,21}	Raw image/ scene + visual filters	Accessibility performance metrics/ User interactions or decisions
Persona-Based Role Playing	Uses models conditioned with persona-based input to emulate user responses and behaviors.	Accessibility assessment ^{23,68} Education ⁶⁷ Synthetic accessibility data generation ⁶⁹	Raw image + Persona description	Simulated assessment Interactive simulated diagnose Simulated interactions/ decisions

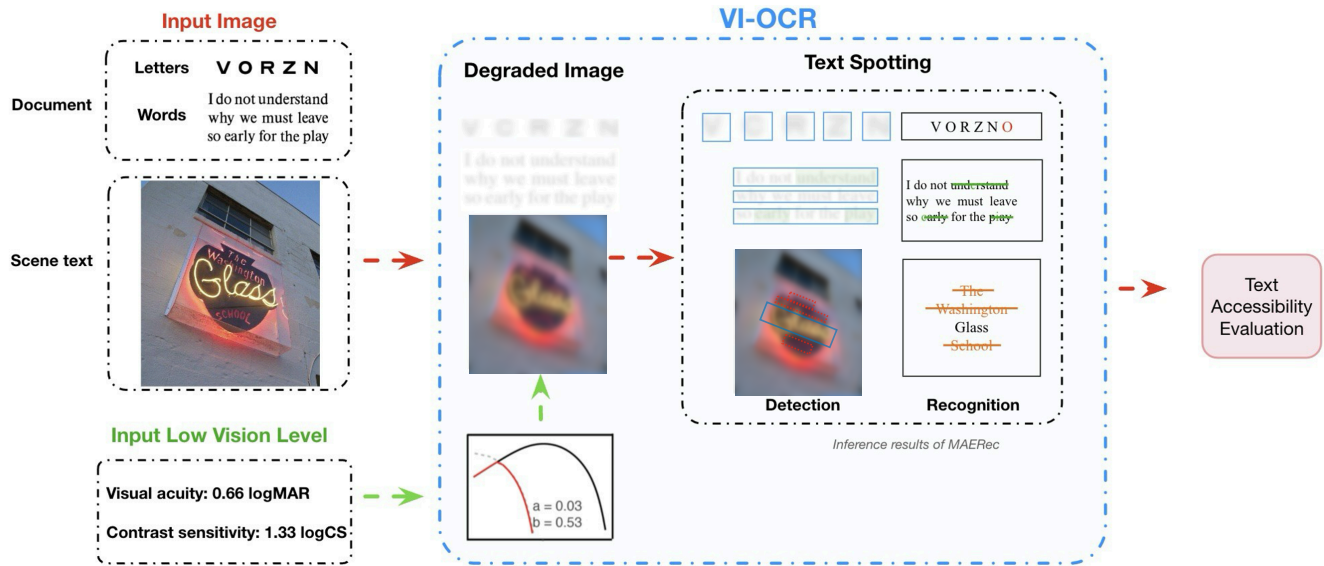


Figure 1. VI-OCR includes image degradation and text spotting modules. The degradation module produces image inputs simulated under varying levels of low vision by adjusting visual acuity and contrast sensitivity. The text recognition module then evaluates reading performance on degraded images to quantify text accessibility under each vision condition. Recognition errors typically include **auto-filled non-existent words** (e.g., recognizing an "O" that does not exist), **character misrecognitions** (e.g., reading "play" as "piay"), and **word omissions** (e.g., missing "the", "Washington", or "school").

Table 2. Low Vision Filter Conditions. VA estimates the high frequency cut-off of the corresponding CSF, indicating the finest detail resolvable, while CS estimates the peak contrast sensitivity of the corresponding CSF, identifying the greatest contrast discernible within the CSF. Across all 15 filters, the acuity change range from 0 to 1.81 logMAR.

Vision Condition	Filter No.	a	b	Acuity Change (logMAR)
Baseline	0	1.000	1.000	0.00
	1	0.250	1.000	0.60
	2	0.125	1.000	0.90
	3	0.063	1.000	1.21
	4	0.031	1.000	1.51
High spatial frequency loss	5	0.016	1.000	1.81
	6	1.000	0.355	0.05
	7	1.000	0.089	0.14
	8	1.000	0.045	0.20
	9	1.000	0.022	0.27
Peak contrast sensitivity loss	10	1.000	0.011	0.37
	11	0.288	0.288	0.60
	12	0.157	0.157	0.90
	13	0.086	0.086	0.97
	14	0.048	0.048	1.51
Combined loss	15	0.027	0.027	1.81

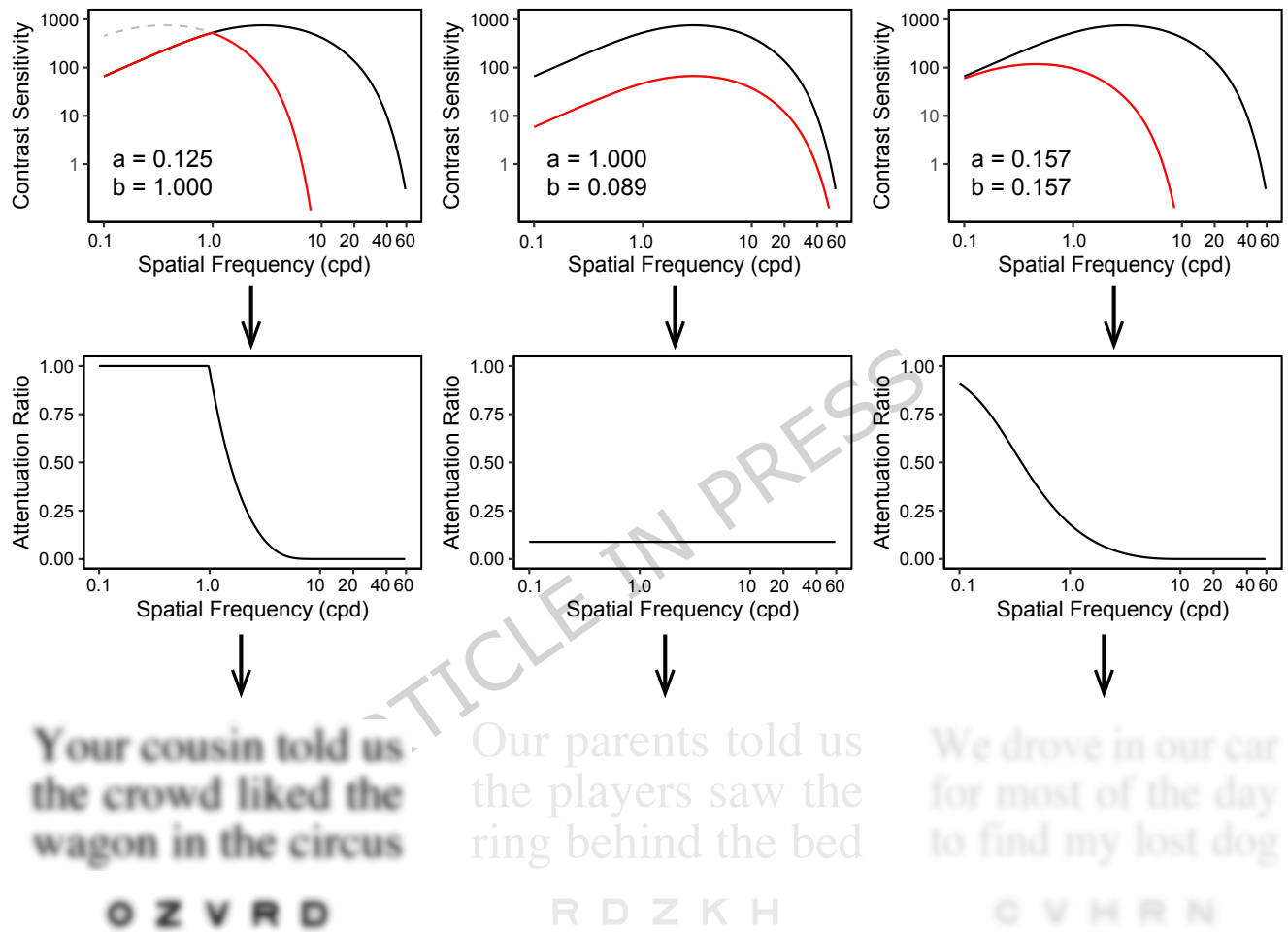
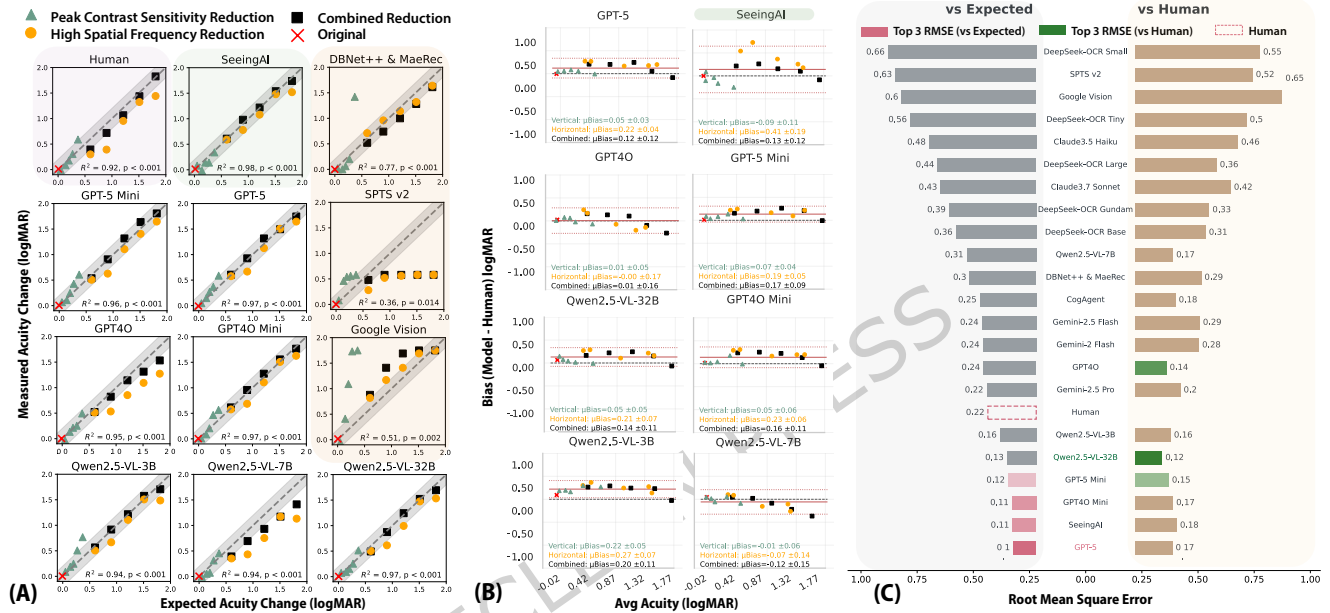


Figure 2. Examples of ETDRS and MNREAD in Experiment 1 after three of the low-vision filters. Left: horizontal shift of CSF that causes high spatial frequency loss. Middle: vertical shift of CSF that primarily simulates peak contrast sensitivity loss. Right: combined horizontal and vertical shift that represent a combination of high frequency and peak contrast sensitivity loss.



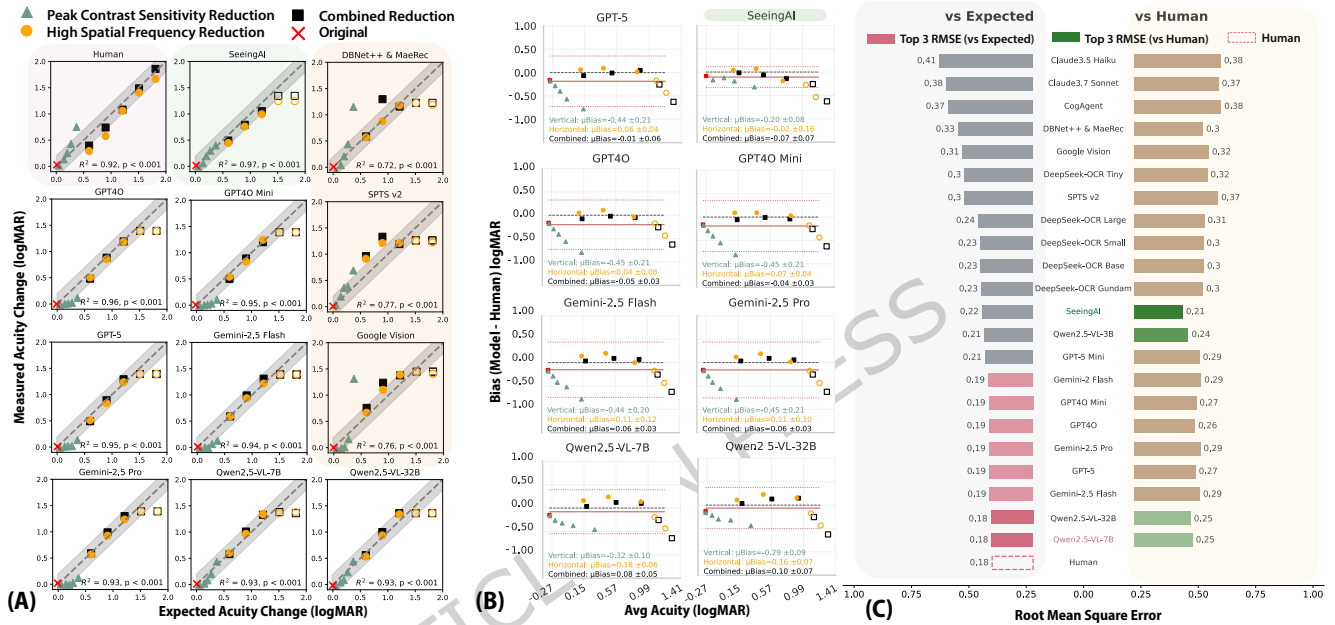


Figure 4. Use VI-OCR to replicate low vision reading acuity: (A) Model and human acuity changes as a function of the expected acuity change for all OCR-specialized models and the top seven VLMs selected based on the highest R^2 values. All presented models show trends consistent with human observers ($R^2 > 0.9$). The gray shaded area indicates the boundary of clinically significant difference (± 0.2 log units). Hollow markers denote acuity levels at which the models reached performance plateaus. (B) The BA plots showing agreement between model and human acuity in logMAR (lower values indicate better vision). The red line denotes the overall limits of agreement (mean bias ± 1.96 SD). Each plot presents average performance and bias under each simulated condition. (C) RMSE comparison. Left: average difference between model/human and expected acuity changes ($RMSE_{exp}$). Right: average difference between model and human acuity changes ($RMSE_{human}$). Red marks the model closest to the expected acuity, and green marks the top three models most closely aligned with human performance. Qwen2.5-vl 7B and 32B both aligned well to expected acuity and human performance.

N = 20 Publicly available images from TotalText



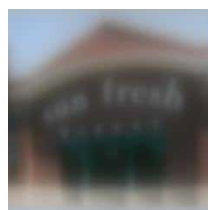
GT: sun fresh MARKET

Prompt: What are all the English words visible in the image?



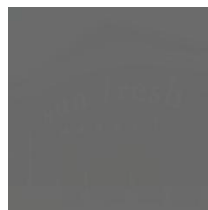
VA: 1.57 CS: 1.49

sun fresh MARKET
 sun fresh MARKET
 sun fresh MARKET



VA: 1.57 CS: 0.59

sus fresh
 fresh
 sun fresh MARKET



VA: 0.13 CS: 0.2

4un
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 sun fresh MARKET

Figure 5. 20 example scene text images in Experiment 2 (Left). As visibility of these images' decreases, the accuracy of human participants declines. In contrast, GPT consistently outperforms under severe vision conditions, indicating a misalignment with human performance (Right). The complete prompt with hint is 'What are all the English words visible in the image? Please output each word in order from left to right and top to bottom. If you cannot recognize it, please leave it blank. Please return strictly in the following format (do not output anything else): ['word1', 'word2', 'word3', ...] Note that the data such as 'word1' above should be replaced with a real word.' All images are selected from the publicly available Total-Text dataset (<https://github.com/cs-chan/Total-Text-Dataset>).

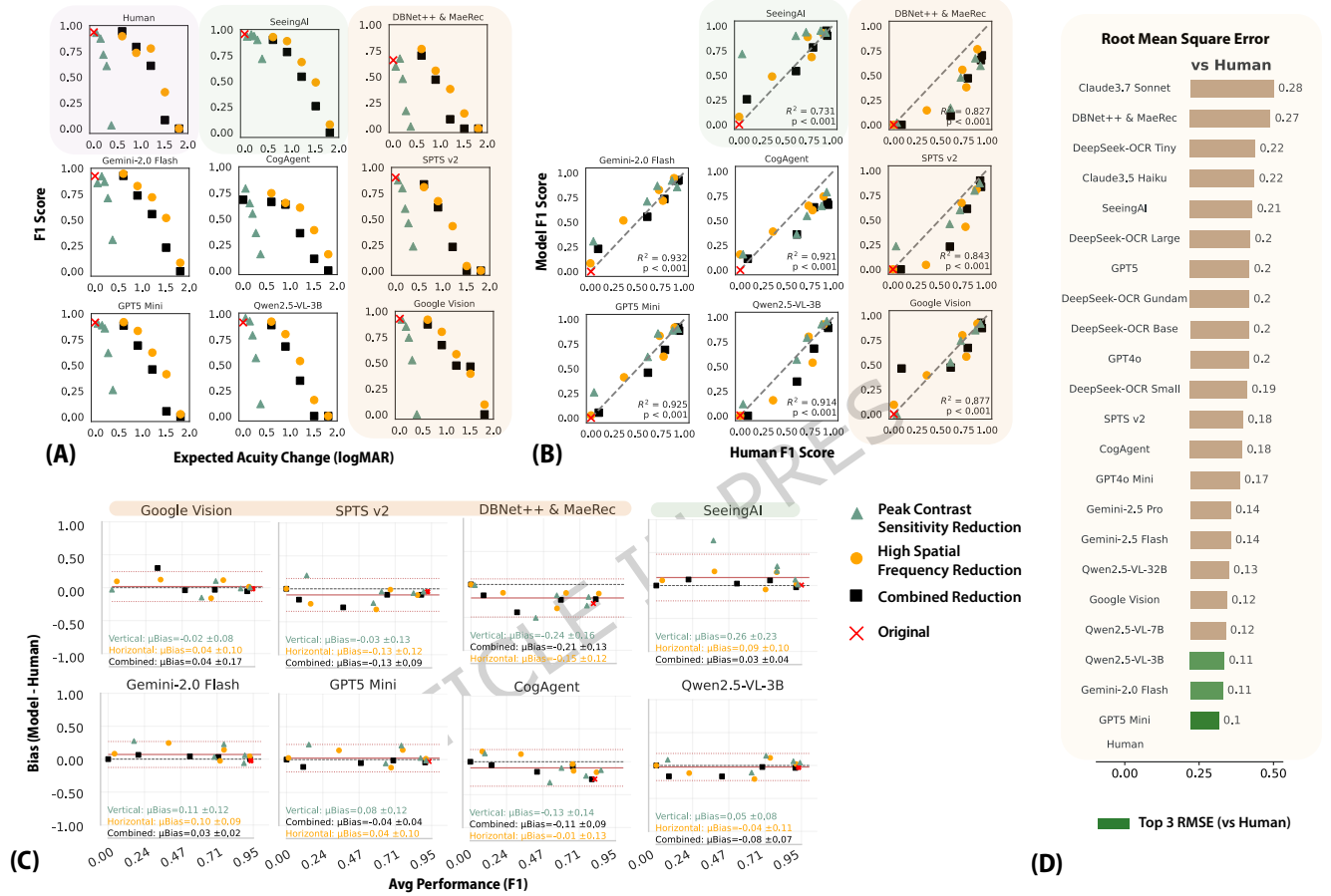


Figure 6. Evaluate VI-OCR performance in scene text recognition: (A) Model and human performance (F1) under each of the 15 low vision simulations. (B) Model and human acuity changes as a function of the expected acuity change. The top four VLMs were selected based on the highest R^2 values, with all showing trends consistent with human observers ($R^2 > 0.9$). (C) The BA plots showing agreement between model and human accuracy. The red line denotes the overall limits of agreement (mean bias ± 1.96 SD). Each plot presents average performance and bias under each simulated condition. (D) Average RMSE between model and human performance. **green** marks the top three models most closely aligned with human performance.

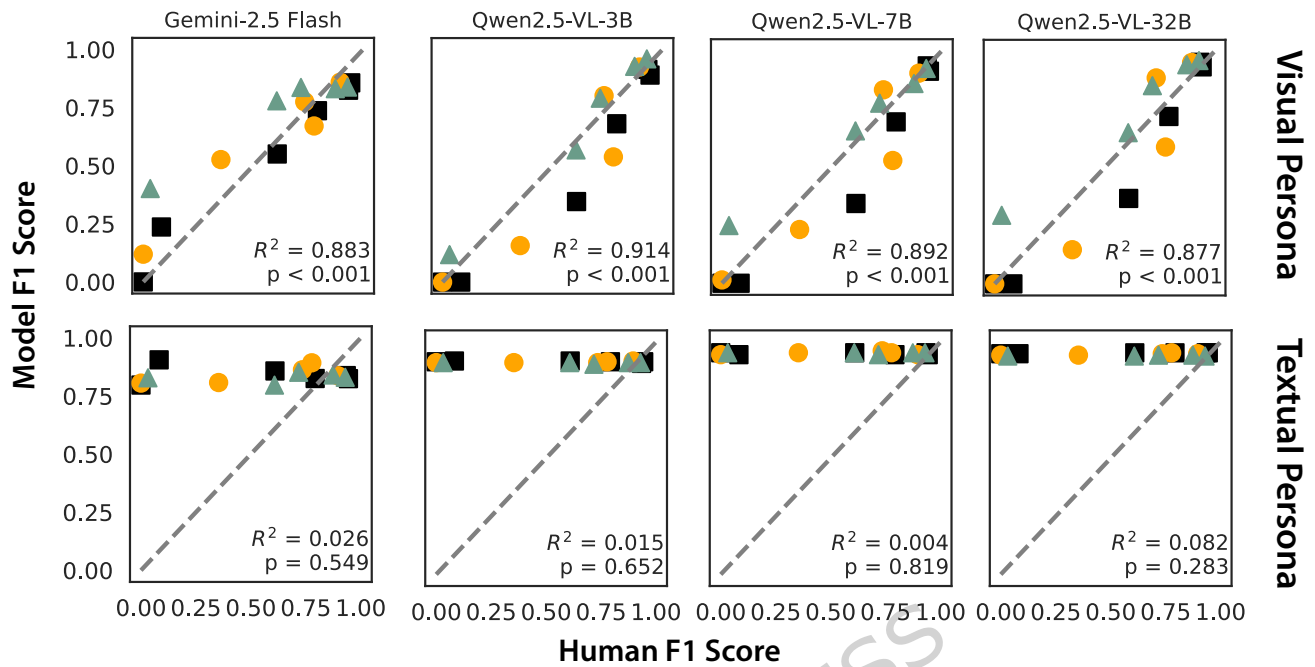


Figure 7. Comparing textual and visual personas in scene-text recognition. The significant overperformance in the textual persona condition revealed models' limited ability to translate textual descriptions into authentic visual experience.

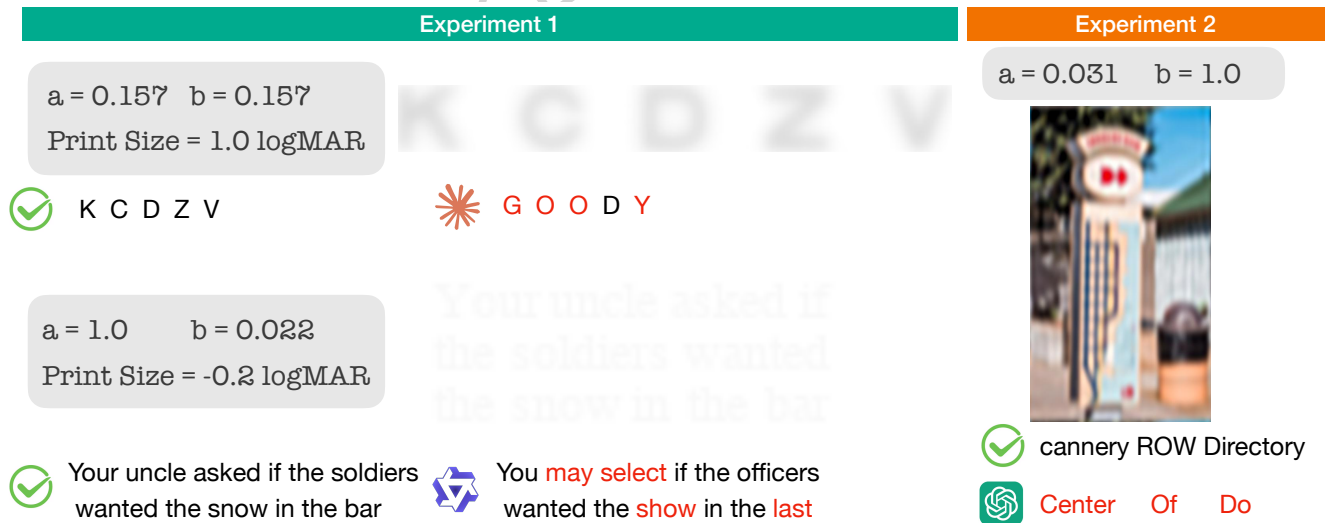


Figure 8. Error cases in letter acuity, word acuity, and scene text recognition tasks due to over-introduction of semantic information. Randomly ordered letters were recognized as familiar words, and low contrast small text was substituted with more frequently occurring words. Note that in these example images, some of the pixelation is due to the enlarged rendering of small print size for visualization (e.g., letters at 1.0 logMAR, words at -0.2 logMAR). In actual tests, the stimuli were shown at their original sizes.

Table 3. F1 Score comparison between human and model. The baseline represents the F1 score under the no-filter condition. Scores for the simulated low vision conditions are the average F1 scores across the five different filters applied under each reduction type. **The highest F1 scores** are highlighted in dark green, **the second-highest** are highlighted in light green. Independent-samples t-tests compared each model with humans under each condition; significance is annotated as * ($p < .05$), ** ($p < .01$), and *** ($p < .001$).

Vision Condition	Open-source	Original	High spatial frequency	Peak contrast sensitivity	Combined
Human		0.934	0.552	0.632	0.486
<i>Vision-Language Models (VLMs)</i>					
GPT4o Mini		0.941	0.580	0.831***	0.481
GPT4o		0.935	0.608	0.854***	0.504
GPT5 Mini		0.917	0.562	0.706	0.417
GPT5		0.953	0.583	0.871***	0.493
Gemini-2.0 Flash		0.920	0.618	0.733*	0.489
Gemini-2.5 Flash		0.825	0.590	0.738	0.476
Gemini-2.5 Pro		0.889	0.637	0.775*	0.509
Claude3.5 Haiku		0.914	0.478	0.459*	0.296*
Claude3.7 Sonnet		0.944	0.406*	0.419**	0.222***
Qwen2.5-VL-3B	✓	0.915	0.485	0.673	0.384
Qwen2.5-VL-7B	✓	0.935	0.502	0.693	0.390
Qwen2.5-VL-32B	✓	0.953	0.515	0.740	0.404
DeepSeek-OCR Tiny	✓	0.784	0.433*	0.393***	0.324
DeepSeek-OCR Small	✓	0.877	0.449	0.460*	0.321
DeepSeek-OCR Base	✓	0.919	0.458	0.452*	0.328
DeepSeek-OCR Large	✓	0.903	0.446	0.452**	0.317
DeepSeek-OCR Gundam	✓	0.928	0.447	0.458*	0.335
CogAgent	✓	0.682	0.509	0.502	0.354
<i>VLM-based App</i>					
SeeingAI		0.952	0.613	0.887***	0.496
<i>OCR-specialized Models</i>					
Google Vision		0.928	0.559	0.608	0.496
DBNet++ & MaeRec	✓	0.656*	0.370**	0.388***	0.253***
SPTS v2	✓	0.901	0.392	0.594	0.336